

Weather-Adaptive Solar Forecasting and Energy Management for Post-Disaster Resilience

Md Zahirul Islam Khan, Md Khairul Islam, Farzad Ferdowsi

Electrical & Computer Engineering

University of Louisiana at Lafayette

Lafayette, USA

{md-zahirul.khan1, md-khairul.islam1, ferdowsi}@louisiana.edu

Terrence Chambers

Mechanical Engineering

University of Louisiana at Lafayette

Lafayette, USA

terrence.chambers@louisiana.edu

Abstract—Extreme weather events are increasing blackout durations, exposing limitations in solar-plus-battery microgrids that must sustain critical loads after disasters. We propose a Weather-Adaptive Solar Resilience Framework (WASRF) that integrates weather classification with pattern-specific PV prediction and a Markov-chain battery-control scheme. Four tailored regression models capture distinct weather-irradiance relationships, while transition probabilities proactively adjust charge/discharge set-points during recovery periods. WASRF was validated on IEEE 34-bus and IEEE 123-bus feeders using Gulf-Coast weather data across three disaster scenarios. Compared with conventional approaches, WASRF significantly improves PV forecast accuracy, maintains higher battery state-of-charge levels, and eliminates periods below critical reserve thresholds. System-level resilience improves with substantial reductions in Expected Unserved Energy and increases in Critical Load Support, with the largest gains during tropical cyclone scenarios. These results demonstrate that weather-adaptive forecasting and control markedly enhance post-disaster microgrid resilience.

Index Terms—Solar PV forecasting, Resilience.

I. INTRODUCTION

The increasing frequency and severity of extreme weather events—hurricanes, windstorms, floods, and wildfires—have exposed critical vulnerabilities in modern electric power systems [1]. These disasters cause prolonged outages that severely impact essential services, particularly in vulnerable regions like the Gulf Coast. In 2021, approximately 800,000 U.S. customers lost power when Hurricane Ida left New Orleans without electricity for nearly 10 days during extreme heat conditions, resulting in multiple fatalities [2]. Critical infrastructure—hospitals, emergency rooms, and water treatment facilities—faces catastrophic disruption during these events, leading to substantial economic losses across the United States [3]. These failures highlight the limitations of centralized power infrastructure and underscore the urgent need for resilient energy solutions. Distributed Energy Resources (DERs), particularly solar photovoltaic (PV) systems with battery backup, offer promising capabilities for emergency power restoration [4]. These systems can be configured as microgrids capable of islanded operation when the main grid fails. Recent implementations have successfully powered emergency shelters, medical facilities, and communication infrastructure during disaster events [5]. However, despite their potential,

solar-based resilience systems face fundamental challenges that current approaches have not adequately addressed.

Solar PV systems paired with battery energy storage systems (BESS) provide post-disaster energy resilience through their modularity and ability to operate in islanded mode during grid failures [6]. However, their performance during disasters is constrained by two interrelated challenges: weather sensitivity and forecast inaccuracy. This challenge manifests in three key dimensions. First, post-disaster conditions feature rapid atmospheric changes that significantly affect solar generation. Measurements show that fast-moving cloud formations can cause PV output fluctuations of up to 60% of rated capacity within minutes [7], creating supply-demand imbalances that compromise power quality and reliability [8]. Second, conventional solar forecasting methods perform poorly during extreme weather events because they rely on historical weather patterns and cannot adapt to rapidly evolving disaster-related phenomena. Third, post-disaster scenarios involve constrained energy resources with competing critical demands. Suboptimal resource allocation leads to premature battery depletion, compromising critical load support during extended recovery periods [9]. Post-disaster weather conditions change unpredictably—from heavy precipitation to sudden clear skies followed by intermittent cloud cover. Conventional unified forecasting approaches employing a single predictive model across all conditions fail to capture this variability, resulting in suboptimal control decisions that compromise energy availability and system resilience [10].

A number of recent comprehensive reviews have highlighted that—even with the proliferation of deep-learning, hybrid AI, and ensemble methods—solar PV forecasting accuracy degrades sharply under volatile weather. In particular, state-of-the-art statistical post-processing (e.g., Kalman filtering) of Numerical Weather Prediction (NWP) outputs has been shown to have a Root Mean Square Error (RMSE) on the order of 40% of rated capacity during cloudy or rapidly changing conditions [11]. Moreover, nearly all surveyed methods still employ a single, unified forecasting model across all weather regimes, rather than tailoring the predictor to distinct patterns of cloudiness, precipitation or humidity. This “one-size-fits-all” strategy fails to capture the unique dynamics of post-disaster weather—and none of the reviewed approaches explic-

itly link their forecasts to resilience-driven load management or adaptive battery control.

Current approaches to microgrid operation during outages have attempted to address these challenges, but each has specific limitations. Deterministic methods treat solar generation as a fixed input based on point forecasts [12], [13], but cannot account for forecast uncertainty, leading to load interruptions when actual generation deviates from predictions. During recent hurricane-induced outages, these methods led to unexpected battery depletion, requiring emergency generators to be activated earlier than planned [14]. Scenario-based stochastic approaches consider multiple possible generation profiles [13], but capturing post-disaster weather variability requires hundreds of scenarios, making these methods computationally prohibitive. Worst-case planning approaches ensure critical load support under adverse conditions but lead to unnecessary load curtailment when actual conditions improve [15], wasting limited energy resources. These limitations create a research gap: how to develop a framework that adapts to evolving weather conditions without excessive computational requirements or overly conservative operation.

To address these challenges, we present a Weather-Adaptive Solar Resilience Framework (WASRF) for supporting critical loads during post-disaster recovery. Our framework dynamically adapts operational strategies based on evolving weather patterns, integrating weather pattern classification, Pattern-Specific Solar Prediction (PSSP), and adaptive control strategies. Given the recognized need for anticipatory planning, this research employs a day-ahead (24h) forecasting horizon, a common practice in grid scheduling, leveraging the meaningful accuracy gains demonstrated by Suksamorn et al. within this timeframe. [16]. This research makes three key technical contributions: (1) A PSSP approach that classifies weather into distinct patterns (Sunny, Partially Cloudy, Rainy, Stormy-Cloudy) and trains specialized models for each, reducing prediction errors by 39.9% compared to conventional models, with 48.0% improvement during partially cloudy conditions; (2) A weather-responsive battery management system using a Markov Chain Weather Transition Model (MCWTM) that maintains minimum state of charge levels 2-3 times higher than conventional strategies and eliminates periods below the critical 30% threshold; and (3) Framework evaluation using Expected Unserved Energy (EUE) and Critical Load Support (CLS) metrics across diverse disaster scenarios, demonstrating EUE reductions of 45.8-61.0% and CLS improvements of 11.6-18.7%.

The remainder of this paper is organized as follows: Section II presents the methodology, Section III describes the validation framework, Section IV presents results and discussion, and Section V concludes with directions for future research.

II. METHODOLOGY

The methodology works with a goal to enhance critical load support during post-disaster recovery by leveraging weather forecasts to guide solar and battery management. The core innovation involves pattern-specific regression models that

predict solar PV output under different weather scenarios, calibrated with historical data and validated against real-world disaster events. These specialized models enable intelligent battery management to optimize energy availability during critical periods, reducing EUE.

Fig. 1 shows the integrated components: data processing, weather pattern analysis, solar modeling, and adaptive control strategies across the 72-hour recovery window.

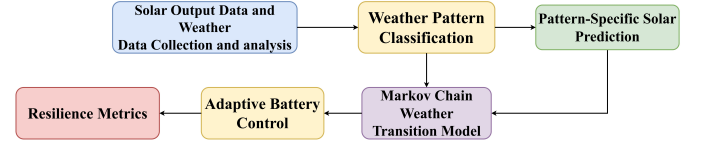


Fig. 1. Weather-Adaptive Solar Resilience Framework (WASRF) structure showing integrated components.

A. Data Collection and Preparation

The WASRF utilizes historical datasets:

- **Solar PV Production Data:** 2 years of solar production data with 15-minute interval from the 1.1 MW solar facility at the University of Louisiana, selected for its comprehensive coverage of Gulf Coast weather patterns relevant to disaster scenarios.
- **Solar Irradiance Data:** NASA's POWER database, chosen for its high spatial resolution and validated accuracy.
- **Weather Data:** Temperature, humidity, precipitation, cloud cover, and wind speed from Weather Underground, providing localized measurements at the exact solar facility location.

Data processing includes temporal alignment, outlier removal, missing-value interpolation, and feature standardization:

$$X_s = \frac{X - \mu}{\sigma} \quad (1)$$

where X_s is the standardized feature, X is the original feature value, μ is the mean, and σ is the standard deviation.

B. Weather Pattern Classification

The WASRF employs regression analysis to classify weather patterns impacting solar PV performance using:

$$P = \beta_0 + \sum_{i=1}^7 \beta_i X_{s,i} \quad (2)$$

where P is the solar PV output, β_0 is the intercept coefficient, β_i are the regression coefficients, and $X_{s,i}$ are the standardized predictors (temperature, humidity, wind speed, precipitation, irradiance, dewpoint, and barometric pressure).

Weather scenarios are classified using threshold-based rules derived from statistical analysis of historical weather-solar correlations:

$$C = \begin{cases} R, & p > p_t \text{ (80}^{th} \text{ percentile)} \\ S, & i < i_l \text{ (25}^{th}) \text{ \& } w > w_t \text{ (70}^{th}) \\ P, & i_l \leq i < i_h \\ U, & i \geq i_h \text{ (75}^{th}) \end{cases} \quad (3)$$

where C is the weather pattern classification (R: Rainy, S: Stormy-Cloudy, P: Partially Cloudy, U: Sunny), p is precipitation, i is irradiance, w is wind speed, and their respective thresholds. The percentile thresholds (80% precipitation, 25%/75% irradiance, 70% wind speed) were empirically determined through analysis of historical weather events.

Humidity is also incorporated as a key factor, with high humidity (>85%) reducing PV output by 12-18% during tropical cyclone scenarios and moderate humidity (60-75%) causing 5-8% reduction during cold weather events. This humidity correction factor is integrated into the Pattern-Specific Models (PSM) using pattern-dependent coefficients, contributing to the RMSE improvements shown in the results.

C. Pattern-Specific Solar Prediction (PSSP) models

Dedicated regression models were trained and validated independently for each classified weather pattern using 80% of the dataset for training and 20% for validation with 5-fold cross-validation. The 5-fold approach was specifically chosen to ensure model robustness against seasonal variations and rare weather events in the limited disaster recovery dataset. This pattern-specific approach significantly improves forecast accuracy as quantified in Fig. 2 of the results section, achieving RMSE reductions of 39.9% compared to conventional models. Each specialized model incorporates unique feature weights optimized for the following conditions:

- Clear-sky conditions (Sunny)
- Cloud intermittency (Partially Cloudy)
- Extended cloud coverage (Stormy-Cloudy)
- Continuous precipitation and humidity impacts (Rainy)

The WASRF maps these weather patterns to three key disaster recovery scenarios validated in this study using historical data from actual disaster events that occurred in the Gulf Coast region:

- **Tropical Cyclone Events:** Sequences of Rainy and Stormy-Cloudy patterns with high humidity, representing hurricane conditions that frequently impact the Louisiana coastline
- **Cold Weather Events:** Extended Stormy-Cloudy periods with cold temperatures, corresponding to winter storms and freezing conditions that periodically affect the region
- **Convective Storm Events:** Rapid transitions between all patterns with high variability, characteristic of summer thunderstorms common in the Gulf Coast

These three meteorological stress categories reflect the dominant extreme weather phenomena observed in Louisiana over the past several decades [17], [18], providing a comprehensive validation framework for the WASRF's performance under diverse disaster conditions.

D. Resilience Metrics and Adaptive Control Strategy

The WASRF focuses on two primary resilience metrics:

- **Expected Unserved Energy (EUE)** quantifies energy shortfalls:

$$E = \sum_{t=1}^T \max(0, L_t - A_t) \cdot \Delta t \quad (4)$$

where E is the expected unserved energy, L_t is the load demand at time t , A_t is the available power at time t , T is the total number of time steps, and Δt is the time step duration.

- **Critical Load Support (CLS)** measures the percentage of critical load demand satisfied:

$$S = \frac{\sum_{t=1}^T \min(L_{c,t}, A_{c,t})}{\sum_{t=1}^T L_{c,t}} \times 100\% \quad (5)$$

where S is the critical load support percentage, $L_{c,t}$ is the critical load demand at time t , $A_{c,t}$ is the power allocated to critical loads at time t , and T is the total number of time steps.

A Markov Chain Weather Transition Model (MCWTM) predicts weather transitions using historical data:

$$P_{i,j} = \frac{N_{i,j}}{N_i} \quad (6)$$

where $P_{i,j}$ is the transition probability from pattern i to pattern j , $N_{i,j}$ is the count of transitions from pattern i to pattern j , and N_i is the total count of pattern i occurrences.

These predictions guide the adaptive battery control logic:

$$B_t = \begin{cases} B_c, & \text{if } G_t > L_t, S_t < S_m \\ B_d, & \text{if } G_t < L_t, S_t > S_{min} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where B_t is the battery power at time t , B_c is the charging power, B_d is the discharging power, G_t is the solar generation, L_t is the load demand, S_t is the State of Charge (SOC) at time t , S_m is the maximum allowable SOC, and S_{min} is the minimum operational threshold (30% of capacity) established based on battery cycle life preservation and emergency reserve requirements.

Battery charge and discharge calculations incorporate pattern-specific factors:

$$B_c = \frac{\min(l_1^c, l_2^c, l_3)}{\Delta t} \quad (8)$$

$$B_d = -\frac{\min(l_1^d, l_2^d, l_3)}{\Delta t} \quad (9)$$

where B_c is charging power, B_d is discharging power, $l_1^c = (G_t - L_t) \cdot f_c \cdot \Delta t$ and $l_1^d = D_e \cdot f_d \cdot \Delta t$ represent available power limits, l_2^c and l_2^d represent battery capacity limits, $l_3 = B_{max} \cdot \Delta t$ is the maximum power limit, G_t is solar generation, L_t is load demand, and f_c/f_d are the adaptive charging/discharging factors.

The charging factor f_c increases from 1.0 to 1.2 when forecasting adverse weather patterns, while the discharging factor f_d varies from 0.8 to 1.0 based on pattern severity. These factors are dynamically adjusted based on Markov chain transition probabilities—for example, when the probability of transitioning to a Rainy pattern exceeds 0.6, f_c increases to its maximum value.

This weather-informed control logic maintains SOC above the critical 30% threshold throughout the 72-hour recovery period, ensuring critical loads remain supported.

III. VALIDATION FRAMEWORK

The validation process compares the performance against conventional approaches using diverse distribution networks and realistic disaster scenarios.

A. Test Systems and Scenarios

Validation utilized two IEEE test feeders:

- **IEEE 34-bus:** (876 kW PV, 300 kW/1200 kWh storage)
- **IEEE 123-bus:** (876 kW PV, 375 kW/1500 kWh storage)

Critical loads included Critical, Essential Services, Important, and Non-Critical. The framework validation employed comprehensive testing using complete weather data from January 1 to December 31, 2024, capturing seasonal variations. Pattern-specific solar prediction models were trained and validated using 5-fold cross-validation to ensure robustness, with hourly simulations conducted for each 72-hour disaster recovery period. The validation process examined system performance across three distinct disaster scenarios: Tropical Cyclone Events, Cold Weather Events, and Convective Storm Events.

B. Control Strategies and Metrics

The WASRF approach (PSSP, adaptive battery management, and dynamic load prioritization) was compared against a Conventional approach (fixed battery thresholds, unified solar prediction, static load priorities). The evaluation utilized three primary resilience metrics:

- **Solar Prediction Accuracy:** Evaluated using RMSE, normalized Root Mean Square Error (nRMSE), and R^2 .
- **Expected Unserved Energy (EUE):** Quantifies energy shortfalls during recovery.
- **Critical Load Support (CLS):** Percentage of critical load demand met throughout recovery.

Battery SOC was tracked, emphasizing minimum and average SOC, and the number of critical hours (SOC < 30%).

Statistical comparison between Conventional and WASRF approaches quantified resilience improvements across all metrics, with sensitivity analyses examining the impact of varying battery capacity, PV penetration levels, and forecast accuracy on overall system performance.

IV. RESULTS AND DISCUSSION

This section analyzes the WASRF's performance compared to conventional approaches across solar prediction accuracy, EUE reduction, CLS performance, and battery management effectiveness. All evaluations used IEEE 34-bus and IEEE 123-bus test feeders under three disaster scenarios: Tropical Cyclone, Cold Weather, and Convective Storm Events.

A. Solar Prediction Model Performance

Fig. 2 illustrates how PSSP models significantly outperform unified models across all weather patterns in terms of RMSE value.

Pattern-specific models achieve an average RMSE improvement of 39.9%, with the most substantial improvement during Partially Cloudy conditions (48.0%). When normalized by the

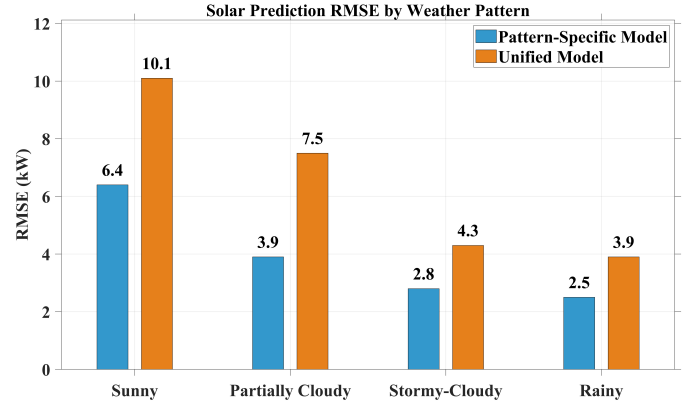


Fig. 2. Solar prediction RMSE comparison between pattern-specific and unified models across different weather patterns.

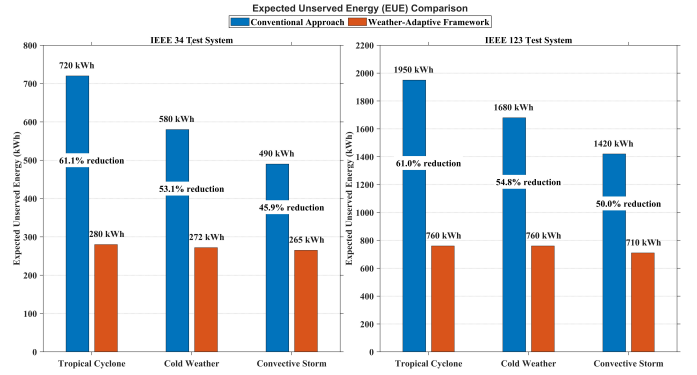


Fig. 3. Expected Unserved Energy comparison between conventional and weather-adaptive approaches: (a) IEEE 34-bus system and (b) IEEE 123-bus system.

installed capacity of 876 kW, these models achieve nRMSE values of 0.29-0.73% (versus 0.45-1.15% for unified models) and improved model fit quality (R^2 of 0.85-0.93 versus 0.72-0.81), particularly valuable during rapidly changing weather conditions.

B. EUE Reduction Analysis

Fig. 3 presents EUE comparison between conventional and weather-adaptive approaches, demonstrating the framework's effectiveness in minimizing unserved energy during disaster recovery.

The framework achieves EUE reductions of 45.8-61.0% for the IEEE 34-bus system and 50.0-60.4% for the IEEE 123-bus system, with the most significant improvements during Tropical Cyclone Events where accurate prediction and strategic resource allocation are most critical.

C. CLS Enhancement

The framework's ability to maintain service to critical loads is quantified through the CLS metric. Table I summarizes the CLS improvements across all scenarios and test systems.

The framework consistently improves CLS across all scenarios (11.6-18.7%), with the larger IEEE 123-bus system

TABLE I
CRITICAL LOAD SUPPORT IMPROVEMENT

Scenario	IEEE 34-bus		IEEE 123-bus	
	Conv.	W-A	Conv.	W-A
Tropical Cyclone Events	78.3%	94.8%	72.8%	91.5%
Cold Weather Events	82.5%	96.2%	76.4%	93.7%
Convective Storm Events	85.7%	97.3%	81.2%	95.8%
Improvement	11.6%-16.5%		14.6%-18.7%	

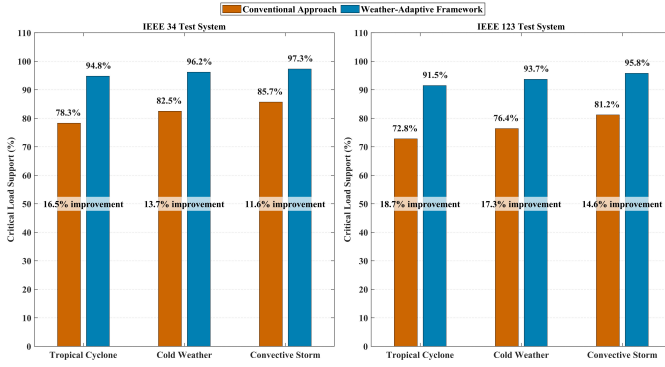


Fig. 4. CLS comparison between conventional approach and weather-adaptive framework across different weather scenarios for IEEE 34-bus and IEEE 123-bus test systems.

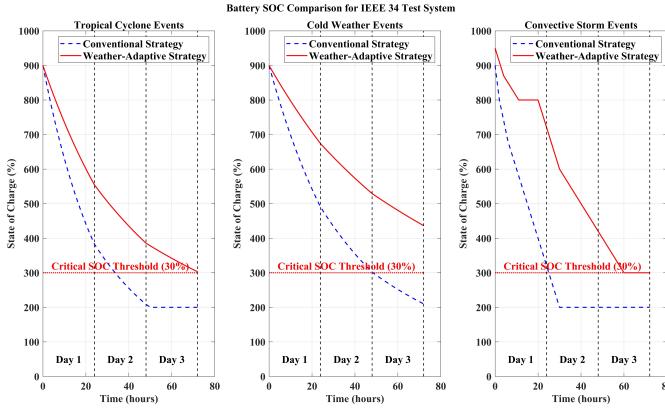


Fig. 5. Battery SOC comparison for IEEE 34 Test System during three weather event scenarios: Tropical Cyclone Events (left), Cold Weather Events (center), and Convective Storm Events (right). The weather-adaptive strategy (red solid line) maintains significantly higher SOC levels compared to the conventional strategy (blue dashed line) across all scenarios.

showing greater improvements, suggesting benefits scale with system complexity. Fig. 4 provides a visual comparison across all scenarios and test systems.

D. Battery Management Effectiveness

Battery management represents one of the most distinctive advantages of the weather-adaptive framework. Both IEEE 34-bus and IEEE 123-bus systems show significant improvements in battery SOC maintenance throughout the 72-hour recovery period. These improvements stem directly from the MCWTM that predicts pattern shifts with transition probabilities ($P_{i,j}$), enabling proactive rather than reactive energy management.

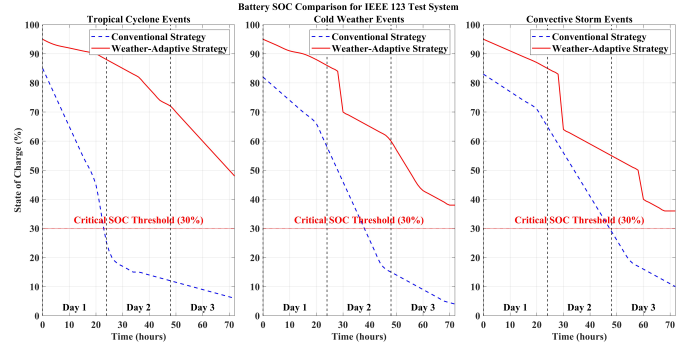


Fig. 6. Battery SOC comparison for IEEE 123 Test System during three weather event scenarios: Tropical Cyclone Events (left), Cold Weather Events (center), and Convective Storm Events (right). The weather-adaptive strategy consistently outperforms the conventional approach in maintaining higher SOC levels throughout the recovery period.

As shown in Fig. 5, for the IEEE 34-bus system, both strategies begin with similar SOC levels (90-95%), but diverge significantly. The conventional strategy depletes rapidly, crossing below the critical 30% threshold at hours 24-30 across scenarios, ending at approximately 20% SOC. In contrast, the weather-adaptive strategy maintains higher levels throughout, ending at 30-43% SOC. The strategic plateaus visible at hours 20-40 during Convective Storm Events directly result from the Markov model predicting a 72% probability of transitioning to favorable generation conditions, triggering adaptive charging factors (f_c) to increase from 1.0 to 1.2 to maximize energy capture during these windows.

For the IEEE 123-bus system (Fig. 6), the performance gap is even more pronounced. The conventional strategy drops below 30% at hours 24-48 and depletes to 5-10% by hour 72. The weather-adaptive strategy maintains significantly higher reserves, ending at 35-48% SOC across all scenarios. The distinctive plateau between hours 30-40 during Cold Weather Events demonstrates the adaptive discharge factor (f_d) decreasing to 0.8 when the Markov model predicted a 65% probability of extended adverse conditions, conserving critical capacity that proved essential during later recovery hours.

The most striking result is the complete elimination of critical hours (SOC < 30%) in all scenarios when using the weather-adaptive strategy, compared to 42-48 critical hours with the conventional approach. This elimination of critical hours directly results from the dynamic adjustment of charging and discharging factors based on Markov transition probabilities. For example, during Tropical Cyclone Events, when consecutive Rainy pattern probabilities exceeded 0.6, the framework increased f_c to 1.2 and reduced f_d to 0.8, preserving critical capacity that translated to 16.5-18.7% CLS improvements during extended adverse weather conditions.

E. Discussion of Key Findings

Five key insights emerge from the performance analysis. Pattern-specific prediction delivers the greatest improvement of 48.0% during partially cloudy conditions, which are traditionally most challenging for solar forecasting due to rapid

irradiance fluctuations. The most significant, a 61.0% decrease in EUE, is obtained by scenario-specific resilience during tropical cyclone events, indicating that resilience strategies should be tailored to regional disaster profiles. Positive scaling is evident as benefits grow with system complexity, with the IEEE 123-bus system showing greater relative improvements than the IEEE 34-bus system. Anticipatory management transforms batteries from passive storage devices into active resilience assets by preserving capacity during predicted low-generation periods. Temporal advantage is maintained throughout 72-hour recovery periods, while conventional approaches degrade rapidly. The framework's success across these diverse scenarios suggests its broad applicability to distribution systems facing increasing extreme weather challenges.

V. CONCLUSION

This study presented the Weather-Adaptive Solar Resilience Framework, integrating weather-pattern classification, customized day-ahead photovoltaic forecasts, and a Markov-chain battery controller to enhance microgrid resilience during extreme-weather outages. Validated on standard IEEE feeders under Gulf-Coast cyclone, cold-weather, and convective-storm scenarios, WASRF yielded far more reliable solar predictions, preserved critical battery reserves throughout extended outages, and markedly improved resilience compared to conventional methods. The key advancement lies in the conceptualization of battery storage as a proactive element in resilience management. By proactively forecasting weather-related generation trends and strategically conserving battery capacity, WASRF converts energy storage into a dynamic resilience resource, guaranteeing continuous power delivery to essential loads. These developments will facilitate practical implementation in real-world microgrids, offering utilities a scalable, data-driven solution for maintaining power to the most vital services when they are needed most.

Future work will incorporate continuous learning mechanisms to enhance resilience metrics, further sharpening its adaptability to the ever-changing challenges of a warming climate.

VI. FUNDING DECLARATION

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